Complaint intake & response operations

Management briefing | Personal portfolio project | Historical snapshot: Dec 2011–Nov 15, 2017

The business task

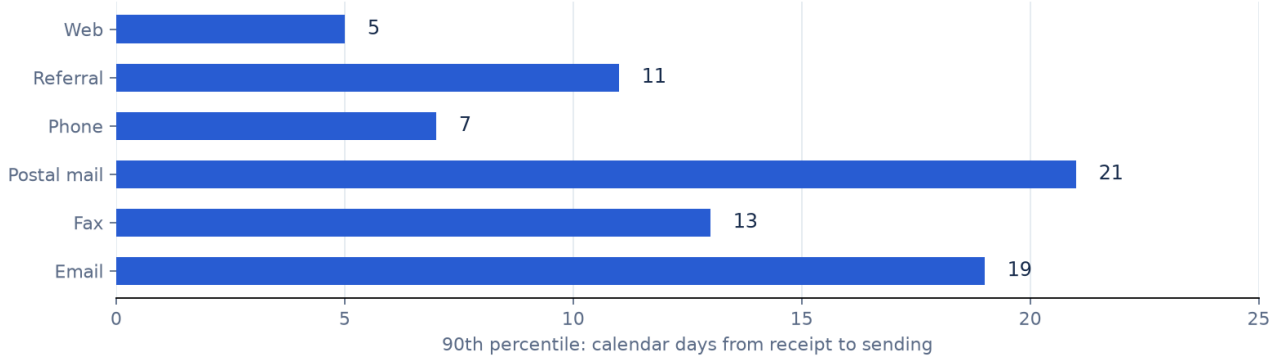
Turn a large complaint intake export into a repeatable operations report, a traceable review queue and an Excel handoff. Preserve reviewer notes across runs and distinguish source response flags from intake routing intervals.

CONTROL / RESULT	OBSERVED VALUE
Source complaint records	903,983
Source timely flag = Yes	879,376 / 903,983 (97.28%)
Valid routing-date intervals	896,931
Median / P90 valid routing interval	0 / 7 calendar days
Negative date intervals	7,052
Intervals above 7-day review threshold	78,562 (8.76% of valid intervals)
Flagged cases, all review reasons	105,969

What the review should focus on

Postal mail has a 21-day P90 routing interval, compared with 5 days for Web. These are descriptive channel observations, with nonnegative intervals only. They do not establish a cause or compare company quality.

The 7,052 negative intervals remain in volume counts and are excluded from duration statistics. Very long positive intervals are also retained and flagged for source verification; an interval alone does not establish how an individual case was handled.



Delivery, controls & repeatability

A practical handoff for remote reporting and data-cleaning work

Run and review

1. Download the documented Kaggle snapshot and run the standard-library Python pipeline. It creates SQLite tables, six SQL extracts, a metrics JSON file and reconciliation results.

2. Open the Excel workbook. The overview reconciles to the channel table; monthly rates use explicit denominators. The review sheet contains 100 longest valid routing-delay examples, with Owner, Review status and Notes entry cells.

3. Export the Review queue sheet as CSV and import its notes by Complaint ID. Untouched template rows do not erase saved notes. Invalid status values and resolved items without an owner and note are rejected before writing.

4. Rerun the report. Saved notes rejoin by stable ID. Notes whose cases are absent from the new source appear in a separate unmatched-notes extract.

Evidence a client can inspect

27 automated tests cover date boundaries, negative intervals, unknown response flags, duplicate/conflicting IDs, note preservation and safe import. Seven full-data controls reconcile row partitions and channel, monthly and product counts. The workbook checks headline values, formula propagation and the resolution-note guard.

Example SQL: duration denominator

```
SELECT channel, COUNT(*) AS valid_intervals,
       AVG(intake_days) AS mean_days
FROM cases
WHERE intake_days >= 0
GROUP BY channel;
```

Scope and limitations

This is a historical portfolio demonstration, not a live backlog. November 2017 is partial. The illustrative seven-day routing review rule is configurable in Python and is not a legal or regulatory SLA. The source timely-response flag is a separate categorical measure. There is no resolution date, market-share denominator or representative sample of all consumers.

Source and privacy

Kaggle: sebastienverpile/consumercomplaintsdata, CC0 Public Domain, originally the CFPB Consumer Complaint Database. Derived exports omit narratives, company names, ZIP codes, state, tags and consent fields. Code is MIT licensed; provenance and a source-file SHA-256 are supplied. Native desktop Excel recalculation was not tested.

Kaggle dataset | CFPB database | Official field definitions